INCH POUND

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27 September 1993
SUPERSEDING
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17 December 1986
(see 6.9)

MILITARY SPECIFICATION

GAUGE, DIFFERENTIAL, DIAL INDICATING

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for differential pressure gauges with analog readout.

1.2 Classification. Differential pressure gauges shall be designated in the following form, as specified (see 6.2):

J	P	L	35	W	1	S	M	5OS2S	M
Range (see 1.2.1)	Units of graduation (see 1.2.2)	Type of scale (see 1.2.3)	Dial size (see 1.2.4)	Dial color (see 1.2.5)	Pressure rating (see 1.2.6)	Sensor material (see 1.2.7)	Fill fluid (see 1.2.8)	Remote sensor system (see 1.2.9)	Gauge housing material (see 1.2.10)

1.2.1 Range. The differential pressure range shall be designated by two characters as specified in 3.3.2.2.3.1. The first character shall be the numerical base value of the range and the second character shall be the multiplier symbol. The multiplier shall be limited to the following:

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AMSC N/A FSC 6685

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and 6.210 inch Hg from 40 to 80 inches

T - multiply by 10
J - multiply by 15
H - multiply by 100
L - multiply by 150
C - compound, multiply by 1
K - compound, multiply by 10
M - compound, multiply by 100

1.2.2 Units of graduation. The units of graduation shall be designated by the corresponding letter symbol and is limited to the following:

P - pounds per square inch (psi)
F - feet of water
I - inches of water
M - inches of mercury
G - gallons per minute
S - special (see 6.2)
N - none (for blank scale)

1.2.3 Type of scale. The type of scale shall be designated by the corresponding letter symbol and is limited to the following:

L - linear scale, for differential pressure and tank level measurements.
S - square root scale, for fluid flow measurements.
N - non-linear, for tank level measurements.
B - blank
S - special (see 6.2)

1.2.4 Dial size. The dial size shall be designated by the corresponding numerical symbols and are limited to the following:

30 - 3 inch circular dial
35 - 3-1/2 inch circular dial
45 - 4-1/2 inch circular dial
60 - 6 inch circular dial

1.2.5 Dial color. The dial color shall be designated by the corresponding letter symbol and is limited to the following:

W - white background dial with black graduations and markings
B - black background dial with white graduations and markings
S - special (see 6.2)

1.2.6 Pressure rating. The pressure rating (in psi) shall be designated by the corresponding numerical symbol and is limited to the following:

1 - 500
2 - 1000
3 - 3000
4 - 6000

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1.2.7 Sensor material. The sensor and remote sensor material shall be designated by the corresponding letter symbols and is limited to the following:

I - inconel
S - corrosion-resisting steel (CRES)
M - monel

1.2.8 Fill fluid. The differential pressure measurement assembly shall be fluid filled. When a remote sensor system is specified, it shall also be fluid filled. Fill fluids shall be based on the temperature range and fluid specific gravity of the tank media to be measured. Fill fluids shall be designated by the corresponding letter symbol and shall be limited to the following:

Fluid class designation	Fill fluid	Temperature range (°F)	Specific gravity	Remarks
B	ethylene glycol (68%) water (32%)	-60 to 220	1.08	Acceptable for oxygen use to 500 psi
D	distilled water	35 to 200	1.00	Oxygen service, hazardous, nuclear and potable water
M	turbine hydraulic oil maxlube #43, grade high viscosity index (Shasta Petroleum), or equal	-60 to 220	0.86	
I	silicon, Dow corning #200-20, or equal	-50 to 300	0.95	Capillary tube 50 feet maximum
S	silicon, Dow corning #550, or equal	-40 to 450		

1.2.9 Remote sensor system. Remote sensor systems shall consist of the remote sensor, a protective and/or sealable housing, and capillary tubing to connect the remote sensor to the differential pressure assembly. The remote sensor system shall be designated by five characters as specified in 1.2.9.1 and 1.2.9.2. When a remote sensor system is not specified, the designation shall be "ZZZZZ".

1.2.9.1 Remote sensor capillary tubing. The remote sensor capillary tubing shall be designated by four characters. The first two shall designate the length of the tubing. The third character shall designate the material of the capillary tubing. The fourth character shall designate the use of one or two capillary tubes. The designators are limited to the following:

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(a) Length: The two characters shall be numerical. They shall be the length of the tubing in feet in increments of 5 feet. The length of the tubing shall be limited to between 5 (designator "05") feet and 95 (designator "95") feet.

(b) Material: The material shall be designated by an alpha character as follows:

C - copper nickel (70/30)
S - corrosion-resisting steel (CRES)

(c) Quantity: The number of tubes used for the remote sensor system shall be designated by a numerical character as follows:

1 - single (one) tube
2 - double (two) tubes

1.2.9.2 Remote sensor housing. The remote sensor housing material shall be designated by a single alpha character and shall be limited to the following:

I - inconel
M - monel
S - corrosion-resisting steel (CRES)

1.2.10 Gauge housing material. The gauge housing material shall be designated by a single alpha character and shall be limited to the following:

S - corrosion-resisting steel (CRES)
B - brass
C - copper-nickel
M - monel

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2)

SPECIFICATIONS

FEDERAL

QQ-A-591 - Aluminum Alloy Die Casting
QQ-N-281 - Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections
QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought (UNS N05500)

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MILITARY

MIL-S-196	- Support Items, Accessories, and Kits, Mechanical; Packaging of.
MIL-S-901	- Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-C-915	- Cable and Cord, Electrical, for Shipboard Use, General Specification for.
MS 3102	- Connector, Receptacle, Electric, Box Mounting, Solder Contacts, AN Type
MS 3106	- Connector, Plug, Electric, Straight, Solder Contacts, AN Type
MIL-C-5015	- Connectors, Electrical, Circular Threaded, AN Type, General Specification for.
MIL-P-5425	- Plastic Sheet, Acrylic, Heat Resistant.
MIL-C-5541	- Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-S-8805	- Switches and Switch Assemblies, Sensitive and Push (Snap Action), General Specification For.
MIL-E-15090	- Enamel, Equipment, Light-Gray (Formula No. 111).
MIL-C-15726	- Copper-Nickel Alloy, Rod, Flat Products (Flat Wire, Strip, Sheet, Bar, and Plate) and Forgings.
MS16142	- Boss, Gasket Seal Straight Thread Tube Fitting, Standard Dimensions for.
MIL-T-16420	- Tube, Copper-Nickel Alloy, Seamless and Welded (Copper Alloy Numbers 715 and 706)
MIL-R-83248	- Rubber, Fluorocarbon Elastomer, High Performance, Fluid, and Compression Set Resistant.
MIL-P-83461	- Packing, Preformed, Petroleum Hydraulic Fluid Resistant, Improved Performance at 275°F (135°C).
MIL-C-85049/1	- Connector Accessories, Electrical, Backshell, Environmental, Cable Sealing, Grounding (Without Strain Relief), Category IC

STANDARDS

MILITARY

MIL-STD-108	- Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-167-1	- Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

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2.2 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR MECHANICAL ENGINEERS (ASME)

B40.1 - Gauges - Pressure Indicating Dial Type - Elastic Element. (DoD adopted)

(Application for copies should be addressed to the American Society for Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 167 - Standard Specification For Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip (DoD adopted)

A 213 - Standard Specification for Seamless Ferritic and Austenitic Alloy-Steel Boiler, Superheater, and Heat-Exchanger Tubes

A 240 - Standard Specification For Heat Resisting Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels. (DoD adopted)

A 276 - Standard Specification for Stainless and Heat -Resisting Steel Bars and Shapes. (DoD adopted)

A 479 - Standard Specification for Stainless and Heat-Resisting Steel Bars and Shapes for Use in Boilers and Other Pressure Vessels

A 564 - Standard Specification for Hot-Rolled and Cold-Finished Age-Hardened Stainless and Heat-Resisting Steel Bars and Shapes

A 581 - Standard Specification for Free-Machining Stainless and Heat-Resisting Steel Wire and Wire Rods. (DoD adopted)

A 582 - Standard Specification for Free-Machining Stainless and Heat-Resisting Steel Bars, Hot-Rolled or Cold Finished. (DoD adopted)

B 21 - Standard Specification for Naval Brass Rod, Bar, and Shapes.

B 26 - Standard Specification for Aluminum-Alloy Sand Castings. (DoD adopted)

B 85 - Standard Specification For Aluminum-Alloy Die Castings. (DoD adopted)

B 124 - Standard Specification for Copper and Copper-Alloy Forging Rod, Bar, and Shapes. (DoD adopted)

B 164 - Standard Specification for Nickel-Copper Alloy Rod, Bar, and Wire

B 194 - Standard Specification for Copper-Beryllium Alloy Plate, Sheet, Strip, and Rolled Bar. (DoD adopted)

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B 209 - Standard Specification for Aluminum and Aluminum -Alloy Sheet and Plate. (DoD adopted)
B 443 - Standard Specification for Nickel-Chromium -Molybdenum-Columbium Alloy (UNS N06625) Plate, Sheet, and Strip.
D 2000 - Standard Classification System for Rubber Products in Automotive Applications
D 3935 - Standard Specification for Polycarbonate (PC) Unfilled and Reinforced Material

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

AMS 5699 - Alloy Wire, Corrosion and Heat Resistant 72 Ni - 15.5 Cr - 0.95 (Cb+Ta) - 2.5 Ti - 0.70 Al - 7.0 Fe Spring Temper, Precipitating Hardenable
AMS 5833 - Alloy Wire, Corrosion and Heat Resistant 20Cr - 15Ni - 40Co - 7.0Mo - 0.06Be - 16Fe Solution Heat Treated and Cold Drawn.
AMS 5844 - Alloy Bars, Corrosion Resistant 20 Cr - 35 Ni - 35 Co - 10 Mo Vacuum Induction Plus Consumable Electrode Vacuum Melted Solution Heat Treated and Work Strengthened

(Application for copies should be addressed to the Society of Automotive Engineers, Engineering Resource for Advancing Mobility, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Nongovernment standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 Materials. Materials of component parts of indicating and sensing assemblies shall be as specified in tables I and II. Cast iron or glass shall not be used.

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TABLE I. Indicating assembly materials.

Part	Material	Material specification	Remarks
Case	Aluminum[1/]	ASTM B 26 ASTM B 85 QQ-A-591	2/
Dial	Aluminum[1/] Corrosion-resisting steel	ASTM B 209 ASTM A 167	See 1.2.3, 1.2.4, 1.2.5
Window	Plastic	MIL-P-5425 ASTM D 3935	Lexan 300 series (or equivalent)
Movement	Corrosion-resisting steel	ASTM A 240 ASTM A 276 (300 or 400 series)	
Pointers	Aluminum corrosion-resisting steel	ASTM B 209 ASTM A 167	
Threaded[3/] fasteners	Corrosion-resisting steel Monel	ASTM A 276, ASTM A 564 ASTM A 581 ASTM A 582 (300 or 400 series) QQ-N-286	
Switches		MIL-S-8805	
Electrical cable		MIL-C-915	
Electrical connector		MIL-C-5015	Both halves See 3.3.2.2.2
Gaskets[3/]	Fluorocarbon rubber (250°F max)	MIL-R-83248	Flat gaskets type 2
O-rings[3/]	Fluorocarbon rubber Nitrile (BUNA N) Rubber (180°F max)	MIL-R-83248 ASTM D 2000 MIL-P-83461	O-rings type 1

see footnotes next page.

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1/ Aluminum shall be protected against corrosion by anodizing in accordance with MIL-C-5541.
2/ Case finish shall include a flash air dry primer with grey enamel in accordance with MIL-E-15090, formula no. 111.
3/ Threaded fasteners, gaskets and O-rings materials shall be used as required in the sensing assembly.

TABLE II. Sensing assembly materials.

Part	Material	Material specification	Remarks
Sensor and remote sensor	Inconel	ASTM B 443	Inconel 625, 718
	CRES	ASTM A 276	CRES 304L, 316L, 321,347
	Monel	ASTM A 240	UNS (S31603)
		QQ-N-286	UNS-N05500
Range springs	Copper-nickel	ASTM B 194	
	Elgiloy	SAE-AMS 5833	
	MP-35N	SAE AMS 5844	
	Inconel	SAE AMS 5699	X750
Housing, gauge	Brass	ASTM B 21	
	CRES	ASTM B 124	
	Copper-nickel	ASTM A 276, ASTM	
	Monel	A 479	
		MIL-C-15726	70-30
		ASTM B 164	
Capillary tubing	Copper-nickel	MIL-T-16420	70-30
	CRES	ASTM A 213	UNS (S31600)
Remote sensor housing	CRES	ASTM A 276, ASTM	CRES 304L, 316L,
	Inconel	A 240	318, 347, UNS
	Monel	QQ-N-281	(S31600)
			CL A or CL B

3.2.1 Recovered materials. Unless otherwise specified herein, all equipment, material and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specified.

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3.3 Construction.

3.3.1 General. The differential pressure gauge shall measure and indicate differential pressure in a manner not to be affected by ambient pressure changes. When specified (see 6.2), the differential pressure gauge shall contain one, two or three mechanically actuated, electrical switches to actuate visual or audio alarms at preset scale readings. The differential pressure gauge shall not be damaged or otherwise affected by overpressure. Provisions shall be provided for zero and span adjustments at the front of the differential pressure gauge.

3.3.2 Parts and subassemblies. Differential pressure gauges shall consist of parts and subassemblies interchangeable and standardized according to each contractor. Parts and subassemblies designated as replaceable parts shall be identified by a code or catalog number system to identify the applicable differential indicator. The differential pressure gauge shall include the following major subassemblies: (a) differential pressure sensing assembly for sensing the pressures and (b) indicating assembly containing the movement, dial, pointers and switches, and switch actuating mechanism as applicable.

3.3.2.1 Differential pressure sensing assembly. The differential pressure sensing assembly shall sense the two input pressures (low and high). It shall convert these pressures to a differential pressure (or signal) to actuate the indicating assembly.

3.3.2.1.1 Differential pressure sensing assembly configuration. The differential pressure sensing assembly size shall be kept to the minimum commensurate with its intended function. The differential pressure assembly shall be configured as specified in figures 1 and 2.

3.3.2.1.2 Connections. The differential pressure sensing assembly shall be provided with two connections, one at the top and one at the bottom of the assembly (see figure 1) for each input (low and high). The connections shall be internally threaded, 7/16-20UNF-2B as specified in MS16142. A solid plug with a rubber O-ring seal shall be provided to fill the unused connection. The plugged connection shall be used for draining or venting the assembly, as applicable.

3.3.2.1.2.1 Remote sensor system connections. When a remote sensor is specified, the capillary tubing used to join the remote sensor to the differential pressure sensing assembly shall be welded or brazed to the differential pressure sensing assembly bottom connections (low and high) and the remote sensor structural housing. The capillary tubing shall be 0.25 inch outside diameter and 0.058 inch wall thickness. The tubing shall be one continuous length between the differential pressure sensing assembly and the remote sensor for lengths less than 40 feet The tubing shall have a maximum of two union type joints, which shall be welded, for lengths greater than or equal to 40 feet.

3.3.2.2 Indicating assembly. The indicating assembly shall include the movement, limit stops, adjustment mechanism, switches, dial, pointers, transparent window, window retaining devices, case and so forth.

3.3.2.2.1 Switches. Electrical switches shall have an electrical rating of 5 amps at 125 volts (V) alternating current (ac) or 2.5 amps at 30 V direct

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current (dc). Unless otherwise specified (see 6.2), switches shall normally be open. Switches shall automatically actuate and remain actuated as the pointer moves to the switch setpoint and beyond. The switches shall be adjustable over 5 percent to 95 percent of the dial scale and shall be field adjustable to the desired set-points without recalibration of the entire differential pressure gauge. Adjustments of the switches shall be from the front of the case.

3.3.2.2.2 Wiring. The interface of the ships wiring with the differential pressure gauge having electrical switches shall be accomplished by use of an electrical connector (receptacle and mating plug) in accordance with MIL-C-5015. The connector shall be provided with the gauge. The receptacle for gauges having two switches shall be a MS3102E14S-6P in accordance with MS 3102. Pins A and D shall be used for the switch common connections. Pins B and E shall be used for the closed contacts which indicate the measured parameter is above the required setpoint value. Pins C and F shall be used for the closed contacts which indicate the measured parameter is below the required setpoint value. The mating plug shall be an MS3106E14S-6S in accordance with MS 3106 with cable clamp as specified in MIL-C-85049/1. For indicators having 3 or 4 switches, the receptacle shall be a MS3102E14S-12P employing the same wiring sequence as specified for a two switch application extended to the additional contacts required. The mating plug shall be a MS3106E14S-12S. The electrical connector port shall be located as specified in figure 3.

3.3.2.2.3 Dials. Unless otherwise specified (see 6.2), dials shall be in accordance with ASME B40.1. Split dials are acceptable for access to internal parts.

3.3.2.2.3.1 Ranges. Unless otherwise specified (see 6.2), ranges and minor graduations used for differential pressure measurement shall be in accordance with table III. Liquid level, fluid flow, and special ranges shall have range designations in accordance with table III and shall be as specified (see 6.2).

TABLE III. Ranges and minor graduations.

Range designation	Differential pressure indicator range	Minor graduations
A	0-20 in H_2O	0.25 in H_2O
B	0-30 in H_2O	0.5 in H_2O
C	0-40 in H_2O	0.5 in H_2O
D	0-50 in H_2O	0.5 in H_2O
E	0-100 in H_2O	1.0 in H_2O
F	0-150 in H_2O	2.0 in H_2O
G	0-200 in H_2O	2.0 in H_2O

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TABLE III. Ranges and minor graduations - Cont'd.

Range designation	Differential pressure indicator range	Minor graduations
H	0-300 in H_2O	5.0 in H_2O
I	0-400 in H_2O	5.0 in H_2O
J	0-15 psi	0.25 psi
K	0-30 psi	0.5 psi
L	0-60 psi	1.0 psi
M	0-100 psi	1.0 psi
N	0-150 psi	2.0 psi
O	0-200 psi	2.0 psi
P	0-300 psi	5.0 psi
Q	0-500 psi	5.0 psi
R	10-0-10 in H_2O	0.25 in H_2O
S	20-0-20 in H_2O	0.5 in H_2O
T	50-0-50 in H_2O	1.0 in H_2O
U	100-0-100 in H_2O	2.0 in H_2O
V	200-0-200 in H_2O	5.0 in H_2O
W	5-0-5 psi	0.1 psi
X	30-0-30 psi	1.0 psi
Y	50-0-50 psi	1.0 psi
Z	100-0-100 psi	2.0 psi
1	Liquid level (see 6.2)	
2	Fluid flow (see 6.2)	
3	Special (see 6.2)	

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3.3.2.2.4 Pointers. Indicating pointers shall be black or white in color (opposite to dial color). The tip of the pointer shall pass within one minor graduation line-width of the inner end of the minor graduations. The pointer shall only overlap the minor graduations. An "index pressure" pointer shall be provided. The index pressure pointer shall be red and adjustable to any pressure on the scale. Pointer tip shall be in accordance with ASME B40.1.

3.3.3 Cleaning. After fabrication, individual parts and subassemblies shall be cleaned of smudges; loose, spattered, or excess solder; weld metal; metal chips and mold release agents or any other foreign material which might detract from the intended operation, function, or appearance of the equipment. (This would include any particles that could loosen or become dislodged during the normal expected life of the equipment.) All corrosive material shall be removed. Whenever possible, this cleaning shall take place before final assembly. All assembled equipment shall be cleaned of contaminants such as lubricating oils, mold release agents, waxes, sand, corrosion products, solder fluxes, finger prints, dust, and so forth. The nature of the contaminant shall be determined to the extent that a suitable cleaning solvent can be selected for item removal. The inertness of the materials of construction to the solvent shall be determined to prevent damage to electrical and mechanical properties. After cleaning, moving parts shall be relubricated and the assembly allowed to dry to remove trapped or soaked cleaning fluid. Cleaning processes shall have no deleterious effect on the equipment or parts. Special cleaning requirements such as cleaning for oxygen service shall be specified (see 6.2).

3.3.4 Mounting. Gauges utilizing a differential pressure assembly less than or equal to the envelope of figure 1 shall be suitable for flush mounting on standard thickness gauge panels. Mounting and panel cut-out dimensions shall be as specified in figure 4. When gauges are equipped with switches, the electrical connector shall not prevent or interfere with flush mounting. Gauges utilizing a differential pressure assembly greater than the envelope of figure 1 but less than or equal to the envelope of figure 2, shall be premounted to a support bracket similar to that shown in figure 5.

3.3.4.1 Remote sensor mounting. When remote sensor systems are specified, the remote sensor housing shall be mounted by means of a suitable support bracket, flange mounting, or direct pipe connection for applications below 50 psi. The support bracket or flange shall be made of the same material as the sensor housing and shall be attached to the sensor housing by welding or bolting.

3.4 Performance. Performance tolerances are specified in percent of gauge span.

3.4.1 Accuracy. The accuracy of differential pressure gauges shall be as specified in 3.4.1.1 through 3.4.1.4 when tested as specified in 4.7.1.

3.4.1.1 Gauges without capillary tube connected remote sensors or mechanically actuated switches. Accuracy of dial indication shall be plus or minus 1 percent.

3.4.1.2 Gauges with mechanically actuated switches. Accuracy of dial indication shall be as follows:

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(a) With switch contacts disengaged plus or minus 1 percent.
(b) With switch contacts engaged plus or minus 1-1/2 percent.
(c) At point of switch actuation plus or minus 3 percent.

3.4.1.2.1 Switch set point. The accuracy of the switch set point shall be plus or minus 1 percent.

3.4.1.3 Accuracy with capillary tube connected remote sensor systems. The accuracy of gauges with capillary tube connected remote sensor systems shall be as specified in 3.4.1.3.1 through 3.4.1.3.2.1.

3.4.1.3.1 Gauges without mechanically actuated switches. Accuracy of the dial indication for gauges with capillary tube connected remote sensor systems and without mechanically actuated switches shall be plus or minus 1-1/2 percent.

3.4.1.3.2 Gauges with mechanically actuated switches. Accuracy of the dial indication for gauges with capillary tube connected remote sensor systems and with mechanically actuated switches shall be as follows:

(a) with switch contacts disengaged plus or minus 1-1/2 percent.
(b) with switch contacts engaged plus or minus 2 percent.
(c) at point of switch actuation plus or minus 3-1/2 percent.

3.4.1.3.2.1 Switch set point. The accuracy of the switch set point with capillary tube connected remote sensor system shall be plus or minus 1-1/2 percent.

3.4.1.4 Friction error. Friction error shall be not greater than 3/4 percent for all ranges. For gauges with mechanically actuated switches, friction error at switch set point shall be not greater than 1-1/2 percent. For gauges with mechanically actuated switches and capillary tube connected remote sensors, friction error at switch set point shall be not greater than 2 percent.

3.4.2 Repeatability. The repeatability of the dial indication shall be within 1/2 percent. The repeatability of the switch set point shall be within 1/2 percent when tested as specified in 4.7.2.

3.4.3 Switch dead band. Switch dead band between make and break shall be not greater than 5 percent when tested as specified in 4.7.3.

3.4.4 Inclination. Maximum deviation of gauge indication and switch set point resulting from inclination shall be not greater than 3 percent when tested as specified in 4.7.5.

3.4.5 Overpressure. During the reference measurements conducted after overpressure, gauge performance shall have no values with deviations greater than the following from the preoverpressure test reference measurements:

1 percent of full scale for gauges with ranges of 60 psid and below.

1.5 percent of full scale per 500 psi applied pressure, not to exceed 10 percent of full scale, for gauges with ranges above 60 psid.

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The gauge shall be adjustable to the accuracy requirements of 3.4.1 at the completion of the tests specified in 4.7.6.

3.4.6 Temperature. After the temperature tests, gauge performance shall meet the requirements specified in 3.4.1.

3.4.6.1 High temperature. Gauges shall be exposed to 145°F ambient without damage. Accuracy of dial indication and switch set point shall be within plus or minus 2 percent when tested as specified in 4.7.7.1.

3.4.6.2 Low temperature. Gauges shall be exposed to 40°F ambient without damage. Accuracy of dial indication and switch set point shall be within plus or minus 2 percent when tested as specified in 4.7.7.2.

3.4.7 Humidity. During the reference measurement conducted during the pre-cycle conditioning period and after the post-cycle conditioning period of the humidity test, the gauge performance shall meet the accuracy requirements of 3.4.1.2. During the accuracy check conducted 18 hours into the second cycle of the humidity test, the accuracy of the gauge indication shall be within plus or minus 2 percent of each pre-cycle reference measurement point, and the accuracy of the switch actuation shall be within plus or minus 4 percent of the pre-cycle reference measurement as indicated by the master test gauge. Temperatures and humidities at which this test is performed shall be within plus or minus 5°F and plus 5 percent, minus 0 percent relative humidity of the values specified on figure 6 when tested as specified in 4.7.8.

3.4.8 Degree of enclosure. The gauge shall be splashproof as specified in MIL-STD-108. There shall be no evidence of water leakage into the gauge when tested as specified in 4.7.9. A reference measurement performed after conclusion of the test shall meet the requirements of 3.4.1.

3.4.9 Salt spray. The gauge shall be physically examined and an accuracy test performed after the conclusion of the salt spray test. The gauge shall show no corrosion or other damage which would adversely affect the operation of the gauge or switches or disassembly of the gauge. Improper operation is also cause for failure. The reference measurement performed after the conclusion of the test and prior to disassembly for inspection shall meet the requirements of 3.4.1 when tested as specified in 4.7.10.

3.4.10 Vibration. Gauges shall withstand the vibration test as specified in 4.7.11. Pointer oscillation shall be not greater than 5 percent peak-to-peak of the applied test pressure at any frequency between 5 to 50 hertz (Hz). Gauges incorporating switches shall be monitored to ensure the switches do not clatter or change state during testing. After the vibration test, gauge performances shall meet the requirements as specified in 3.4.1.

3.4.11 Shock resistance. Gauges shall withstand the shock test as specified in 4.7.12. There shall be no evidence of damage other than a shift in dial indication which shall be not greater than 2 percent for any single blow or 3 percent for the nine blows. After the shock test, gauge performance shall meet the requirements as specified in 3.4.1 with adjustments.

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3.4.12 Pressure cycling. Gauges shall withstand not less than 260,000 pressure cycles as specified in 4.7.13 without damage. After the cycling test, gauge performance shall meet the requirements as specified in 3.4.1 with adjustments.

3.4.13 Switch setpoint range. Switches shall adjust from 5 to 95 percent of gauge range and meet the requirements of 3.4.1, 3.4.2 and 3.4.3, respectively, when tested as specified in 4.7.14.

3.5 Technical data. (see 6.3 and appendix).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Inspection system. When specified (see 6.2 and 6.3).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article shall be performed and passed prior to production. First article inspection shall be performed on samples which have been produced with equipment and procedures normally used in production. First article inspection shall consist of the examinations and tests specified in table IV.

4.3.1 Test sample size. Two samples of each configuration shall be subjected to first article inspection.

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4.3.2 Order of inspection. The sample gauges shall be subjected to the inspections specified in table IV in the order listed. Any deviation in the test order shall be first approved by NAVSEA.

TABLE IV. First article inspection.

Inspection or test	Requirement paragraph	First article inspection paragraph
General examination	3.2 and 3.3	4.5
Accuracy	3.4.1	4.7.1 and 4.7.15
Repeatability	3.4.2	4.7.2
Switch dead band	3.4.3	4.7.3
Inclination	3.4.4	4.7.5
Overpressure	3.4.5	4.7.6
Temperature	3.4.6	4.7.7
Humidity	3.4.7	4.7.8
Degree of enclosure	3.4.8	4.7.9
Salt spray	3.4.9	4.7.10
Vibration	3.4.10	4.7.11
Shock	3.4.11	4.7.12
Pressure cycling	3.4.12	4.7.13
Switch setpoint variation	3.4.13	4.7.14

4.4 Quality conformance inspection. Each gauge offered for delivery shall be subjected to the quality conformance inspections specified in table V.

TABLE V. Quality conformance inspection.

Inspection or test	Requirement paragraph	Quality conformance inspection paragraph
General examination	3.2 and 3.3	4.5
Accuracy	3.4.1	4.7.1
Repeatability	3.4.2	4.7.2
Switch dead band	3.4.3	4.7.3

4.4.1 Noncompliance. If a gauge fails to pass the quality conformance inspections, the manufacturer shall notify the procuring activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the procuring activity has been taken. After the corrective action

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has been taken the quality conformance inspection shall be repeated. Final acceptance and shipment shall be withheld until the quality conformance inspections have shown that the corrective action was successful.

4.5 General examination. Each gauge shall be given a thorough examination to determine conformance to the requirements of this specification with respect to materials, finish, workmanship, construction, assembly, dimensions and markings.

4.6 Test conditions and test procedures.

4.6.1 Test conditions. Except where the following factors are the variables, the tests specified in 4.7 shall be conducted with the equipment operating under the following conditions:

(a) Ambient temperature shall be 77 ± 3°F.
(b) Relative humidity shall be 50 ± 20 percent.

4.7 Tests.

4.7.1 Accuracy. When calibrating a gauge with a remote sensor system, the calibration pressure shall be applied to the remote sensor housing. If the gauge has both high pressure (HP) and low pressure (LP) remote sensors (classification quantity "2" of 1.2.9.1), the gauge may be calibrated with the remote sensors separated by an amount specified for the required calibration, or they may be tested with the remote sensors at the same level with the appropriate pressures applied to simulate calibration at the required separation. For units without remote sensors, calibration pressure is to be applied to the HP side with the LP side vented to atmosphere. For units with remote sensors, the calibration pressure is to be applied as required to simulate the calibration at the required separation. The gauge shall be flexed over its full pres-sure range by slowly increasing and decreasing the applied pressure three times. A calibration measurement shall be made at five equally spaced intervals over the entire range both upscale and downscale. Readings shall be taken both before and after lightly tapping the center of the gauge dial in accordance with ASME B40.1. The error in pressure indication after tapping both upscale and downscale, above the first 5 percent of the scale, shall be not greater than the requirements of 3.4.1 for each test.

4.7.1.1 Friction error. A calibration measurement shall be made of not less than five equally spaced intervals over the entire range both upscale and down-scale. Readings shall be taken both before and after lightly tapping the center of the gauge dial. The difference in readings before tapping and after tapping is the friction error and shall meet the requirements of 3.4.1.3.

4.7.1.2 Switch setpoint. Setpoints shall be adjusted to 25 and 75 percent of full scale. The accuracy of the switch at each setpoint shall be checked during the accuracy calibration and during each reference measurement using a master test gauge (not the gauge under test) and shall meet the requirements of 3.4.1.2.1.

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4.7.2 Repeatability. The calibration measurement specified in 4.7.1 shall be applied three successive times to determine repeatability. Repeatability shall meet the requirements of 3.4.2.

4.7.3 Switch dead band. The switch dead band shall be determined by slowly increasing (or decreasing) the applied pressure until the switch actuates. The applied pressure shall then be slowly decreased (or increased) until the switch deactivates (resets). The switch dead band (or differential) is the difference between the applied pressure for actuation and the applied pressure for deactuation of the switch. The switch dead band shall be the maximum value determined after three consecutive actuation/deactuation cycles and shall meet the requirements of 3.4.3. The normally open/normally closed contacts of single pole, double throw switches shall be monitored simultaneously to verify complete and instantaneous contact transfer at the switch set and reset points. Switch dead band shall be determined at each setpoint tested.

4.7.4 Reference measurement. The following series of checks, constituting a reference measurement, shall be performed when specified in an individual test:

(a) A one trial calibration with not less than five equally spaced intervals over the entire gauge range both upscale and downscale.
(b) Friction error at each interval both upscale and downscale.
(c) Accuracy of set point of each switch.
(d) Switch set point repeatability.

4.7.5 Inclination. The gauge pressure housings shall be filled with water to simulate operational conditions. The low and high pressure chambers shall be filled from separate reservoirs. The water level in these reservoirs shall be maintained constant throughout the test. Calibration pressures shall be applied to the HP reservoir. The gauge shall be inclined for a period of not less than 1 minute in each of the following positions:

(a) 60 degrees forward.
(b) 60 degrees backward.
(c) 60 degrees to the left.
(d) 60 degrees to the right.

In each position a reference measurement (see 4.7.4) shall be made. Performance shall conform to 3.4.4.

4.7.6 Overpressure. Prior to the overpressure test a reference measurement (see 4.7.4) shall be made. The gauge shall successfully withstand the full rated pressure applied for a period of 1 hour to the high pressure (HP) port with the low pressure (LP) port open to the atmosphere. Within 10 minutes of the release of the rated pressure, a reference measurement shall be made for comparison. Performance shall conform to 3.4.5. The full rated pressure shall then be applied to the low pressure port for a period of 1 hour with the high pressure port open to the atmosphere. Within 10 minutes of the release of the rated pressure, a reference measurement shall be made for comparison. Performance shall conform to 3.4.5.

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4.7.7 Temperature.

4.7.7.1 High temperature. The gauge shall be subjected to an ambient temperature of 145°F for a period of not less than 6 hours. A reference measurement shall be made at the end of the 6 hour period. Performance shall conform to 3.4.6.1.

4.7.7.2 Low temperature. The gauge shall be subjected to an ambient temperature of 40°F for a period of not less than 6 hours. A reference measurement shall be made at the end of the 6 hour period. Performance shall conform to 3.4.6.2.

4.7.7.3 Temperature test reference measurement. After completion of the temperature tests, but within 6 hours, a reference measurement shall be made using test conditions of 4.7.1. Gauge performance shall conform to 3.4.6.3 with adjustments.

4.7.8 Humidity. The gauge shall be subjected to four cycles at 24 hours per cycle at the temperature and humidity conditions shown on figure 6. A 4 hour pre-cycle conditioning period shall be performed before the cycling sequence is initiated. Figure 6 shows the 4 hour pre-cycle conditioning period along with one complete temperature and humidity cycle. A post-cycle conditioning period, at the conditions of 77°F and 50 percent relative humidity for a duration of not less than 20 hours, shall be performed after the completion of the fourth cycle. A reference measurement shall be performed after three hours of the pre-cycle conditioning period. An accuracy check shall be performed after 18 hours of the second cycle of the test has been completed. A second reference measurement shall be performed after the post-cycle conditioning period. The reference measurements and accuracy check shall be performed in accordance with 3.4.7. Adjustments to the gauge are permitted during the reference measurement after the post-cycling conditioning period only.

4.7.8.1 Humidity cycle requirements. The humidity test pre-cycle and post-cycle conditioning and cycle requirements shall be as follows:

Pre-cycle conditioning:
3 hours (77°F, 50 percent relative humidity)
Reference measurement - shall meet the requirements of 3.4.1.2.
1 hour (77°F, 50 percent relative humidity)

Cycle 1:
1 hour - Raise temperature and relative humidity
18 hours (149°F, 95 percent relative humidity)
1 hour - Drop temperature
4 hours (77°F, 95 percent relative humidity)

Cycle 2:
1 hour - Raise temperature
18 hours - (149°F, 95 percent relative humidity)
Accuracy check: Gauge indication: plus or minus 2 percent deviation from pre-cycle reference measurement values.
Switch actuation: plus or minus 4 percent deviation from pre-

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cycle reference measurement values.
1 hour - Drop temperature
4 hours (77°F, 95 percent relative humidity)

Cycle 3:
1 hour - Raise temperature
18 hours (149°F, 95 percent relative humidity)
1 hour - Drop temperature
4 hours (77°F, 95 percent relative humidity)

Cycle 4:
1 hour - Raise temperature
18 hours - (149°F, 95 percent relative humidity)
1 hour - Drop temperature
4 hours (77°F, 95 percent relative humidity)

Post-cycle conditioning:
20 - 24 hours (77°F, 50 percent relative humidity)
Reference measurement - shall meet the requirements of 3.4.1.2.
Gauge adjustments are allowed during this reference measurement only.

4.7.9 Degree of enclosure. The gauge shall be subjected to a splashproof test specified in MIL-STD-108. Performance shall conform to 3.4.8.

4.7.10 Salt spray. The gauge shall be subjected to a salt spray test in accordance with method 101 of MIL-STD-202 using a 5 percent salt solution. Duration of the test shall be 96 hours. Pressure connections shall be plugged for gauges not designed to operate in seawater. A reference measurement shall be performed after this test. Performance shall conform to 3.4.9.

4.7.11 Vibration. This gauge shall be tested as specified in MIL-STD-167-1. The gauge shall be pressurized to approximately the mid-scale position and the span of pointer oscillation shall be determined at each frequency. Pointer oscillation shall be not greater than the limits as specified in 3.4.10. Gauge with remote sensor systems shall be tested as a complete system. Switches shall be monitored for changes in contact state. After the vibration test, a reference measurement without adjustment shall meet the requirements as specified in 3.4.1.

4.7.11.1 Failure. Resonant frequencies shall be determined by exploratory variations over the frequency ranges and amplitudes. Where resonance frequencies do not occur, the endurance test shall be run at 50 Hz. As a result of the test, no part of the gauge shall become loosened, dislodged or broken and the switches shall not exhibit any chatter or change of state. A reference measurement without adjustment shall meet the requirements as specified in 3.4.1. In addition, pointer oscillation at resonance shall be not greater than 5 percent of applied pressure and center of oscillation shall be within the tolerance as specified in 3.4.1.

4.7.12 Shock test (high-impact). The shock test shall be conducted as specified in MIL-S-901 for grade A, type A class 1 using the appropriate adapters. The gauge shall be pressurized to approximately mid-scale position. Gauges with

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remote sensor systems shall be tested separately. The gauge shall be mounted with a 6D-1 fixture and the remote sensor system shall be mounted with a 4A fixture. Switches shall be monitored for change in contact state. A reference measurement shall be performed prior to the shock test and readings shall be taken after each blow. Following the shock test, gauge performance shall conform to 3.4.11.

4.7.13 Pressure cycling. The gauge shall be tested on a system applying a sinusoidal pressure change for a total of 260,000 cycles. The rate of cycling shall be approximately 1 cycle per second. The pressure shall be cycled between 40 percent and 60 percent of gauge span. When gauges with remote sensor systems are cycled the pressure shall be applied to the HP housing. Gauge performance shall conform to 3.4.12.

4.7.14 Switch setpoint variation. Accuracy (see 4.7.1), repeatability (see 4.7.2), and switch dead band (see 4.7.3) shall be determined at individual switch setpoints of 10, 50, and 90 percent of gauge range. Switches shall meet the requirements of 3.4.1.2.1, 3.4.2, and 3.4.3 respectively.

4.7.15 Accuracy with full rated pressure. The accuracy test specified in 4.7.1 shall be repeated with full rated working pressure applied to both LP and HP gauge connections or remote sensors. The calibration pressures shall then be applied to only the LP side, by reducing the pressure on the LP side by an amount necessary to achieve the required calibration pressure. At no point during the test shall the pressure applied to either the LP or HP side exceed the full rated working pressure. The error in pressure indication after tapping both upscale and downscale, above the first 5 percent of the scale shall not exceed the requirements of 3.4.1 for each test.

4.8 Inspection of packaging. Sample packages and the inspection of the packaging (preservation-packaging, packing and marking) for shipment, stowage, and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.5).

5.1 Packaging requirements. The packaging (preservation, packing, and marking) requirements shall be in accordance with MIL-S-196, for the level of preservation (A, B, C, or commercial), the level of packing (A, B, C, or commercial), and marking including other packaging acquisitioning therein, as specified (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. Differential pressure gauges are used in various applications aboard Naval ships. The most prominent of these applications are described in 6.1.1 through 6.1.3.

6.1.1 Differential pressure gauges. Differential pressure gauges are intended to provide an indication of differential pressures for shipboard applications ranging from 20 inches water column to 500 pounds per square inch.

6.1.2 Liquid level gauges. Liquid level gauges employing remote sensor systems are intended for use on fluid level storage tanks where the level to be measured is greater than or equal to 0 to 30 inches water column. Tank capacity curves should be furnished as well as the location of the sensing points for determining the range and calibration of the tank level gauges (see 6.2).

6.1.3 Fluid flow rate indicators. Fluid flow rate gauges are used with orifice, venturi, pitot tube or elbow primary elements. The gauge should have a differential pressure range based on the primary element calibration curve with the applicable fluid (see 6.2). The readout display dial should include calibration pressure equivalent to the rate-of-flow graduated scale.

6.2 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Classification (see 1.2).
(c) Units of graduation for designation "S" (see 1.2.2).
(d) Type of scale for designation "S" (see 1.2.3)
(e) Dial color for designation "S" (see 1.2.5).
(f) Issue of DODISS to be cited in solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.2).
(g) If first article is required (see 3.1).
(h) The number of mechanically actuated electrical switches (see 3.3.1).
(i) Special cleaning requirements, if required (see 3.3.3).
(j) Electrical switches normal position (see 3.3.2.2.1).
(k) Special ranges for designation "XX" (see 3.3.2.2.3.1).
(l) Fluid flow range designation "FF" (see 3.3.2.2.3.1).
(m) Liquid level range designation "LL" (see 3.3.2.2.3.1).
(n) If an inspection system is to be provided (see 4.1.2).
(o) Packaging requirements (see 5.1).
(p) Tank capacity curve, if applicable (see 6.1.2).
(q) Primary element identification and calibration curve, if applicable (see 6.1.3).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/ provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the

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data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference paragraph	DID Number	DID Title	Option
3.4.10, 4.7.11	UDI-23762	Report, vibration testing	MIL-STD-167-1
3.4.11, 4.7.12	DI-ENVR-80708	Shock test report	MIL-S-901
3.5	DI-DRPR-80651	Drawing, engineering	---
4.1.2	DI-QCIC-81110	Plan, inspection and test	---

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L., Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Sub-contracted material and parts. The packaging or preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.6 Provisioning. Provisioning Technical Documentation (PTD), spare parts and repair parts should be furnished as specified in the contract.

6.6.1 Spare parts. When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.7 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals

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are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.8 Subject term (key word) listing.

Fluid flow rate
Liquid level
Pressure
Shock resistance
Switches
Vibration

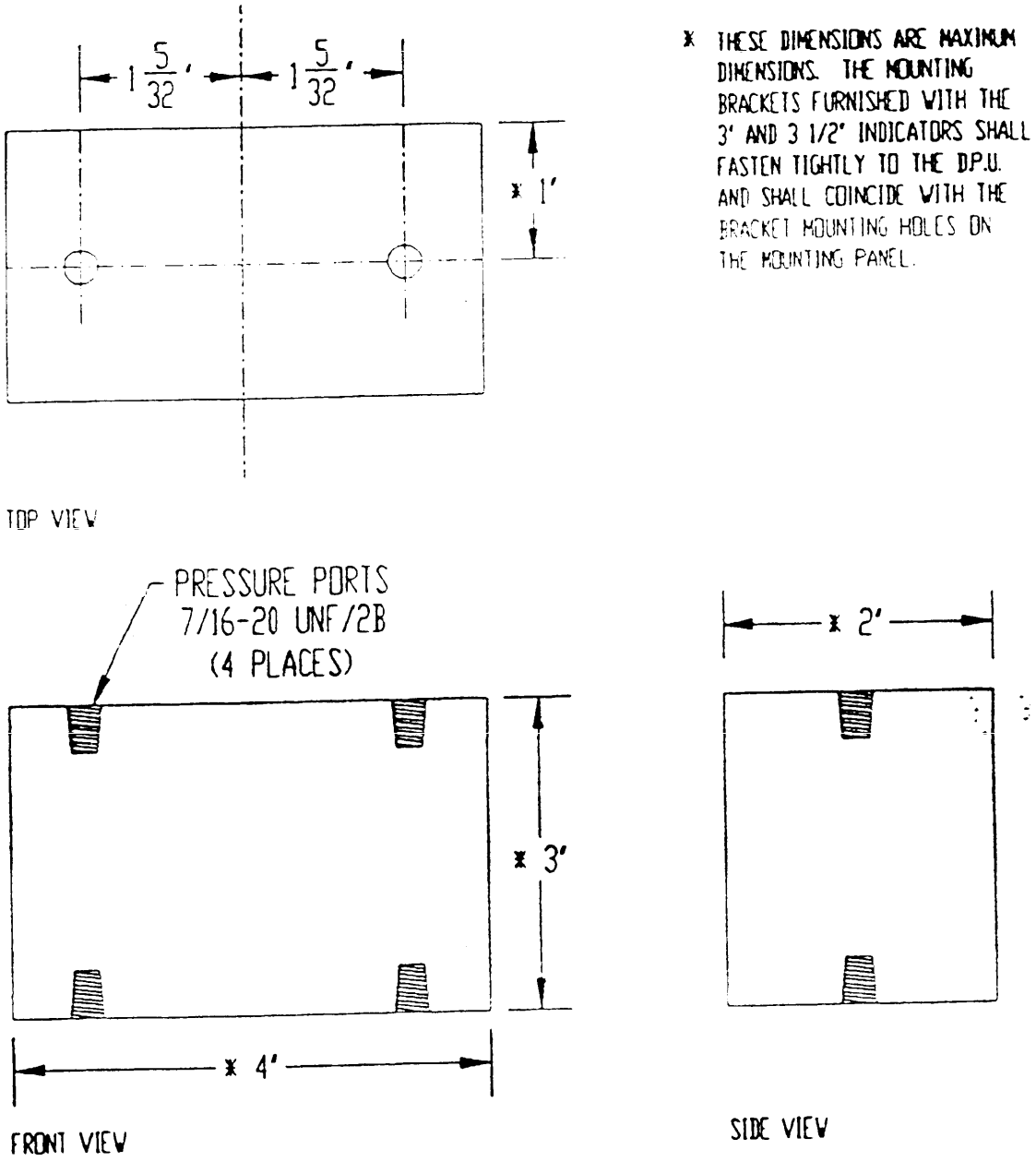
6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 6685-N730)

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DIFFERENTIAL PRESSURE UNIT DIMENSIONS

THE SAME DIFFERENTIAL PRESSURE UNIT IS USED FOR THE 3", 3 1/2", 4 1/2", AND 6" INDICATORS.
DIMENSION TOLERANCE IS 1/8"; HOLE TOLERANCE IS 1/32".



* THESE DIMENSIONS ARE MAXIMUM DIMENSIONS. THE MOUNTING BRACKETS FURNISHED WITH THE 3" AND 3 1/2" INDICATORS SHALL FASTEN TIGHTLY TO THE D.P.U. AND SHALL COINCIDE WITH THE BRACKET MOUNTING HOLES ON THE MOUNTING PANEL.

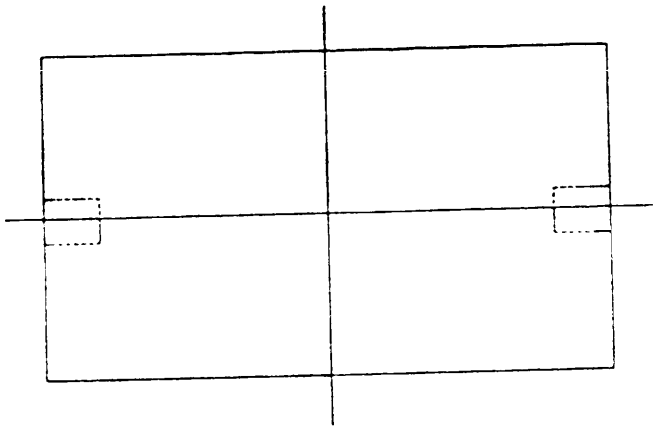
FIGURE 1. Differential pressure unit dimensions.

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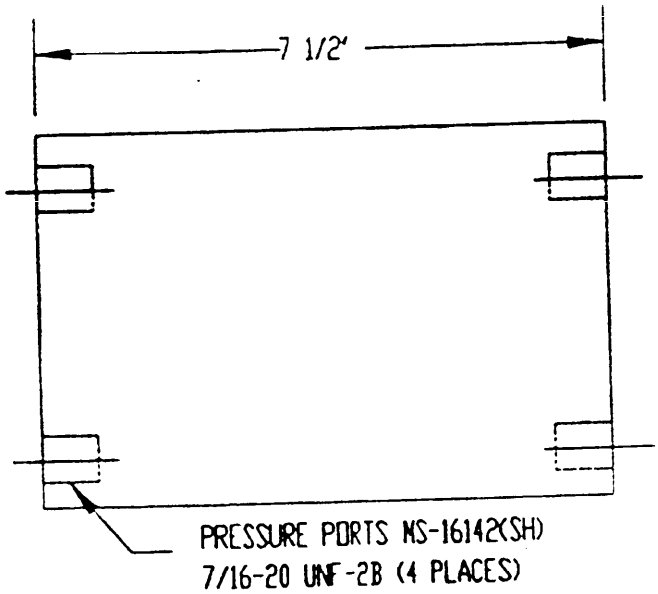
DIFFERENTIAL PRESSURE UNIT DIMENSIONS

THE SAME DIFFERENTIAL PRESSURE UNIT IS USED FOR THE 3", 3 1/2", 4 1/2", AND 6" INDICATORS.
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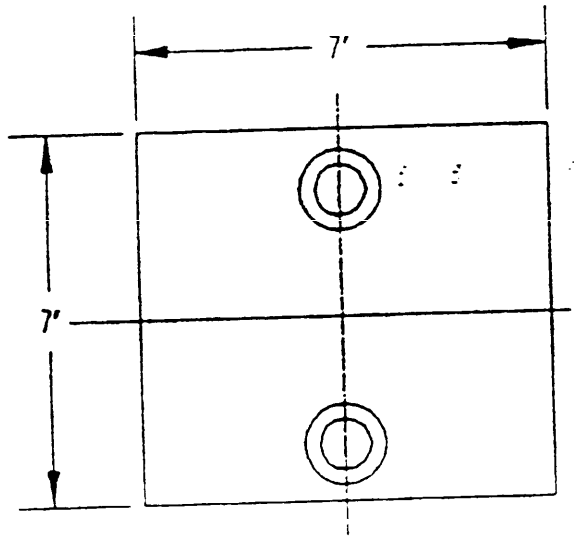
* THESE DIMENSIONS ARE MAXIMUM DIMENSIONS. IF THE DIFFERENTIAL PRESSURE CHAMBERS ARE ROUND THEY SHALL NOT EXCEED 7" IN DIAMETER.



TOP VIEW



FRONT VIEW



SIDE VIEW

FIGURE 2. Low range differential pressure unit.

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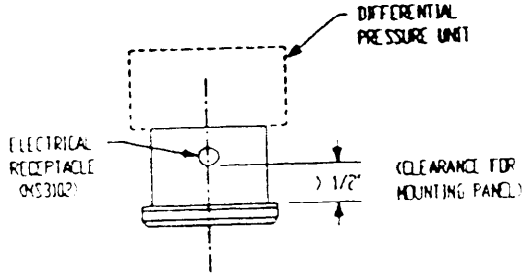
SWITCH WIRING PORT LOCATIONS AND DIMENSIONS

THE WIRING PORTS MUST BE HERMETICALLY SEALED USING STUFFING TUBES.
DIMENSION TOLERANCE IS 1/8"; HOLE TOLERANCE IS 1/32".

3" INDICATOR

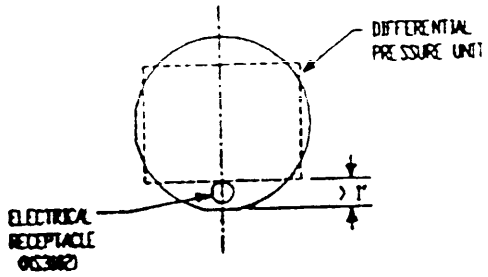
THE 3" INDICATOR DOES NOT CONTAIN SWITCHES.

3 1/2" INDICATOR



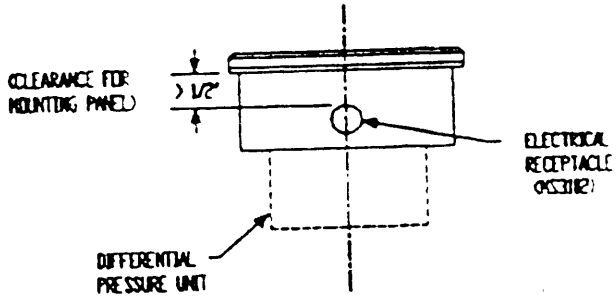
TOP VIEW

4 1/2" INDICATOR



REAR VIEW

6" INDICATOR



BOTTOM VIEW

FIGURE 3. Switch wiring port locations and dimensions.

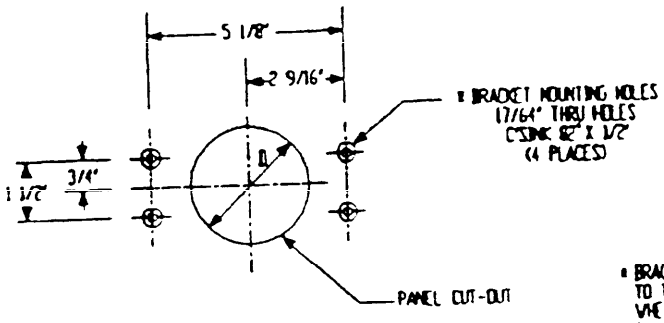
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MOUNTING DIMENSIONS

THE TOLERANCE ON ALL MOUNTING DIMENSIONS IS ± 1/32"

3" INDICATOR

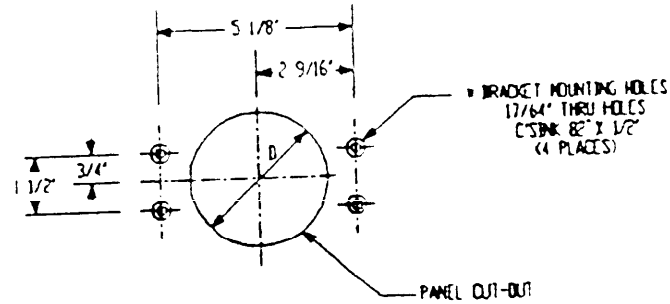
D = 3 9/64"



* BRACKETS SHALL FASTEN SECURELY TO THE I.P. UNIT AND TO THE PANEL WHERE INDICATED. BRACKET MATERIAL SHALL BE CORROSION RESISTANT STEEL

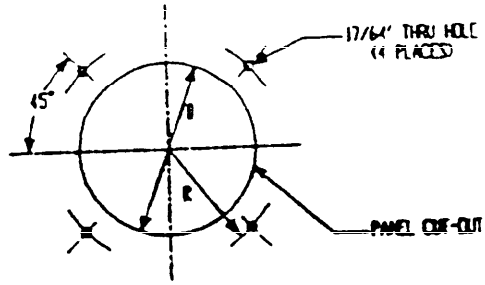
3 1/2" INDICATOR

D = 3 63/64"



4 1/2" INDICATOR

D = 5 15/32"
R = 3 1/16" BOLT CIRCLE



6" INDICATOR

D = 6 1/2"
R = 3 1/2" BOLT CIRCLE

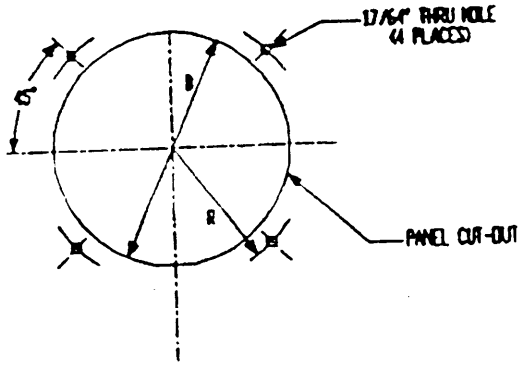


FIGURE 4. Mounting dimensions.

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DIFFERENTIAL PRESSURE UNIT DIMENSIONS

THE SAME DIFFERENTIAL PRESSURE UNIT IS USED FOR THE 3", 3 1/2", 4 1/2", AND 6" INDICATORS.
DIMENSION TOLERANCE IS 1/8"; HOLE TOLERANCE IS 1/32".

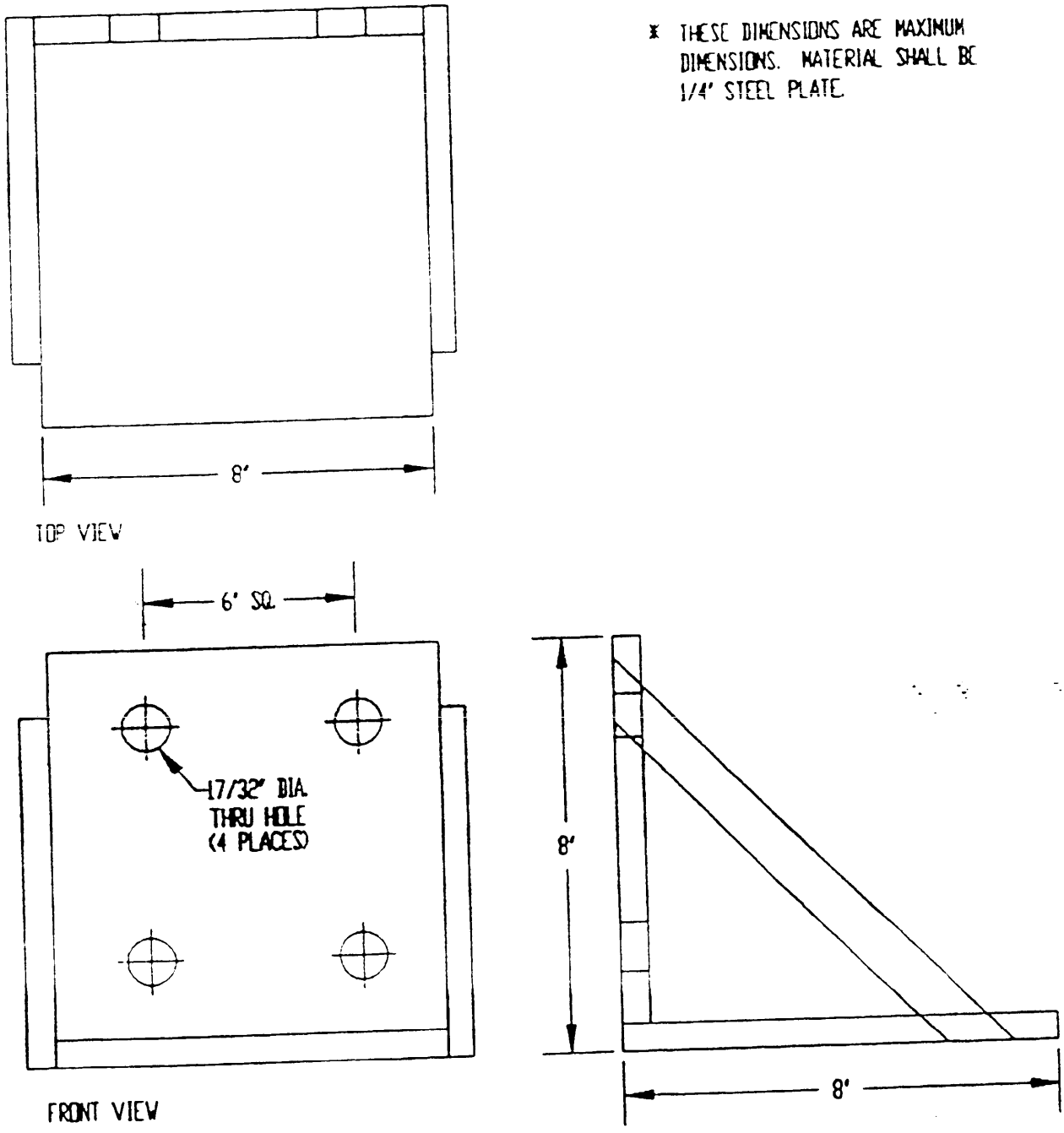


FIGURE 5. Low range differential pressure assembly mounting bracket.

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** Reference measurement is performed

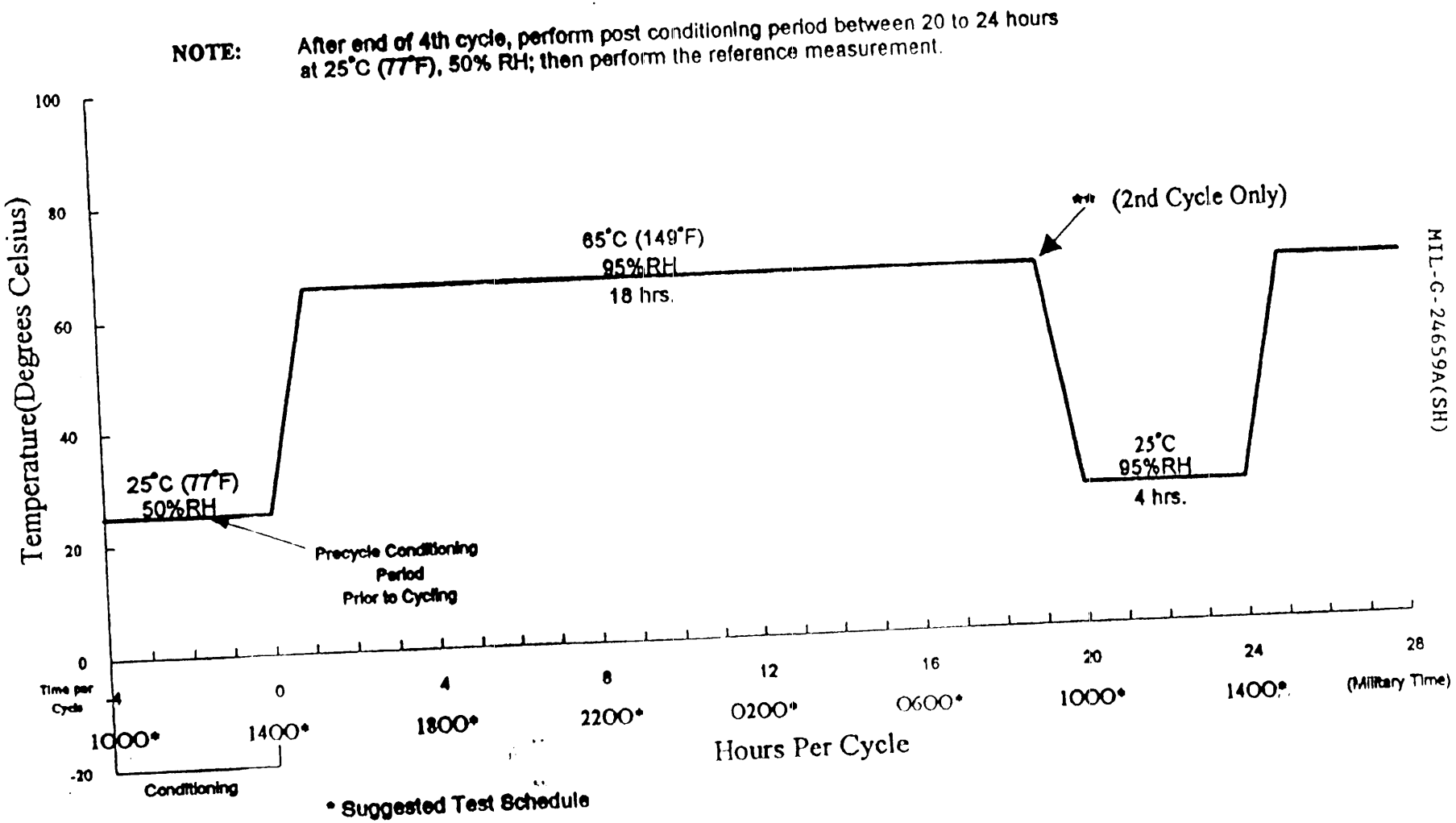


FIGURE 6. Humidity test.

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APPENDIX A

Engineering Drawings Technical Content Requirements

10. Scope

10.1 Scope. This appendix contains the format and content preparation instructions for the development of and revision to a conformance verification drawing. It is not intended that each requirement contained herein should be applied to every type of instrumentation. Portions of this appendix are subject to deletion tailoring depending upon the material, construction and principal of operating requirements that are specified in the individual instrumentation specification or acquisition document. This appendix is mandatory only when data item description DI-DRPR-80651 is cited on the DD Form 1423.

20. APPLICABLE DOCUMENTS.

20.1 Government documents. The following documents form a part of this appendix to the extent specified herein.

SPECIFICATION

MILITARY
MIL-I-45208 - Inspection System Requirements

STANDARD

MILITARY
MIL-STD-17-1 - Mechanical Symbols (Other than Aeronautical, Aerospacecraft and Spacecraft Use) Part 1

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

20.2 Other publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the document which is DoD adopted is that listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

Y 14.1 - Drawing Sheet Size and Format
IEEE 200 - Standard Reference Designations for Electrical and Electronics Parts and Equipment

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018).

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(Nongovernment standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

30. CONFORMANCE VERIFICATION DRAWING FOR INSTRUMENTATION

30.1 Purpose. The conformance verification drawing contains the information necessary to verify that the instrumentation meets the requirements specified in the applicable instrumentation specification and acquisition document.

30.2 Content and format. One conformance verification drawing shall be developed for each specific type of instrumentation. It shall include all ranges, sizes, connections, and other variations. The conformance verification drawing shall include the following minimum information (except as specified in 10.1) and shall be developed to the following format:

30.3 Descriptive data.

a. Instrument identification numbering system for instrumentation. This numbering system shall include, but may not be restricted to, _____ the classification variables.

b. Instrumentation identification number system for replaceable parts.

c. Size, operating data, ranges, scale markings, and other data for proper selection.

d. Test approval data, presenting the following information in tabular form:

(1) Specification classification (unique alphanumeric variables to designate instruments).

(2) Test report number and date.

(3) Facility where test was conducted.

(4) Authorized government activity (NAVSEA) approval letter and date.

e. A statement that instrumentation is in accordance with the requirements of the applicable instrumentation specification of acquisition document and to referenced specifications.

f. Conformance verification drawing acceptance data, presenting the following information in tabular form:

(1) Authorized government activity (NAVSEA) acceptance letter and date.

(2) Revision number.

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30.4 Details of construction.

a. Two or more representative assembly views, as required, to show clearly the details of the design, construction, and assembly of the instrumentation and to identify each part and its location. Identification of parts shall correspond to the list of materials. Assembly shall show how all mechanical parts are joined or attached.

b. Sectional view or notes as necessary to show internal details.

c. Details such as entrance provisions, gaskets, fastening techniques, welding symbols, mounting requirements, and other details as applicable.

d. Module enclosure requirements for separately mounted modules.

e. Dimensions required to ensure interchangeability.

f. Schematic of indicating system.

g. Description of the nature and purpose of any adjustments.

h. Welding procedures with acceptance data, including acceptance letter, report numbers and dates.

i. Any special features.

j. Complete weight.

k. Location, size, and type of connections.

l. Identification of quality control documents which show conformance with MIL-I-45208 of the quality control specification that is listed in the applicable instrumentation specification or acquisition document. The acceptance letter shall be referenced.

30.5 Parts list. The following information shall be presented in tabular form:

a. Item number (corresponding to flag number identifying the part on the conformance verification drawing).

b. Quantity of each part required per assembly.

c. Name of part with sufficient information to readily identify the part (e.g., screws: thread size, length and type of head, shall be specified).

d. Material of part.

e. Material specification (military, federal, or commercial specification number or government activity (NAVSEA) drawing number).

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NOTE: When substitution of a material specification is made, it is the responsibility of the contractor to provide written documentation to substantiate that the submitted material is equivalent to the specified material.

f. Type, class, grade, size, military designation, or other classification of any referenced specification.

g. Part number or identification assigned by assembly supplier.

h. Name of actual manufacturer of part (when applicable).

i. Part number or identification assigned by part supplier (when applicable).

j. Onboard repair parts. Parts that are appropriate for, or are supplied as, onboard repair parts shall be indicated by a symbol in this column.

k. Remarks column. Finishes, platings, or coatings along with the applicable specification or other requirements should be specified in this column.

30.6 Table for special tools. Special tools required for the instrumentation shall be presented in the following tabular form:

a. Item number (corresponding to flag number identifying the tool on the conformance verification drawing).

b. Quantity of each tool required per instrument.

c. Name (description) of tool (including generic name).

d. Tool specification (military, federal, or commercial specification number or government activity (NAVSEA) drawing number).

e. Tool number of identification assigned by assembly supplier.

f. Name of actual manufacturer of tool (when applicable).

g. Tool number or identification assigned by tool supplier (when applicable).

h. Description of tool's application.

i. Remarks column. Special techniques or other usage requirements should be explained in this column.

30.7 Fluid and electrical system schematics. Fluid (piping) and electrical schematics shall be included in the verification conformance drawing when applicable. A complete schematic shall contain all the parts in the fluid or electrical systems. When confusion over its function could result, a simplified schematic containing only the major components shall also be provided. This simplified schematic may be presented in block diagram format.

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a. Complete fluid system schematic. A single schematic shall present clearly the operation and functions of the fluid system within the instrumentation. This schematic shall contain all parts (including valves, fittings, hoses, and tubing) which make up the piping or fluid system. The following features shall be incorporated into the schematic:

(1) A simplified, clear schematic in preference to one showing the physical placement or parts.

(2) A thin, broken line shall be used to represent the boundaries of each subassembly.

(3) Arrows adjacent to the parts shall indicate direction of flow.

(4) Each part shall be identified by the schematic part designation. The numbers shall be assigned in a logical sequence observing glow paths through the system.

(5) All inlet and outlet ports shall be identified.

(6) The methods of attachment for each part shall be designated by an appropriate label or symbol.

(7) The symbol used to designate a part shall be in accordance with MIL-STD-17-1 when applicable.

(8) A brief description of the fluid system operation shall be provided.

(9) Maximum operating pressure of the system and maximum pressure drop through the system (when operating at maximum pressure) shall be specified.

(10) Table for pressure rating and connection of fluid system parts. Supplementary information on parts found in the fluid (piping) system schematic shall be contained in tabular form. One column shall contain the schematic part designation for each fluid system part; e.g., valve V-1 and other similar designations. The other columns shall contain the following information:

(a) Item number

(b) Description

(c) Type (size)

(d) Pressure ratings (including operating, proof, and burst).

(e) Method of attachment, including, as applicable, brazing or welding procedure, bonding agent, and seal (gasket, O-ring, etc.)

(f) End connections, fittings, and adapters.

(g) Relief valve ratings (including cracking and set pressure).

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b. Complete electrical system schematic. A single schematic shall represent clearly the operation and the function of the electrical circuitry within the instrumentation. The schematic shall contain all parts (including components, connectors, alarms, etc.) which make up the electrical system. The following features shall be incorporated into the schematic:

(1) In preparation of the schematic, emphasis shall be placed on simplicity and ease of understanding of circuit operation. Physical placement of components and connecting wiring may be ignored in the interest of simplicity and clarity of this diagram.

(2) A thin, broken line shall be used to represent the boundaries of each unit or subassembly. Terminals, to which external connections are made, shall be shown within these boundaries, with the numbers, markings, type of signal, power and ground, as appropriate.

(3) Each part (such as resistors, capacitors, relays, etc.) shall be given a unique reference designation consisting of a letter denoting the type of part (as required by ANSI/IEEE 200) and a number assigned consecutively. The numbers shall be assigned in a logical sequence of electrical current or signal flow through the circuit.

(4) In addition to where the reference designation parts not conforming to a military specification, where permitted, shall have the following information noted adjacent to the part. Where numerical values are given, a code shall be noted to designate the units used:

(a) Resistors - Resistance, power rating, and tolerances. If variable, an arrow to indicate clockwise rotation of the control shaft.

(b) Capacitors - Capacitance, voltage rating and tolerance.

(c) Reactors - Inductance and voltage rating

(d) Semi-conductors - Type number (JEDDC number is adequate).

(e) Integrated circuits - Operational symbol diagram of input-output relationship, terminal numbering corresponding to a representative schematic and type number.

(5) Supply voltages, phases, and frequencies and transformer terminal voltage shall be indicated and labeled as to purpose.

(6) Table for troubleshooting of electrical system. The content table shall include each test point, as identified on the electrical system schematic, with voltage, waveform or other electrical parameter that should be measured at each test point.

30.8 Selection and installation considerations.

a. Performance data

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(1) Accuracy (in percent of span)

(2) Shock and vibration classification

(3) Degree of water tightness of the enclosure

(4) Electromagnetic interference and pulse susceptibility.

b. Dimensional outline of the instrumentation showing overall and principal dimensions in sufficient detail to establish space requirements in all directions necessary for installation, servicing, exclusive of space required for operator observation of the indication.

c. Special considerations which may affect selection or installation

(1) Ambient temperature range

(2) Calibration points and adjustments

(3) Orientation

(4) Location of instrumentation relative to vibrating equipment

(5) Protection of the instrumentation from pulsations and spikes in the parameter being measured

(6) Selection of the instrumentation range relative to the operating range of the system

(7) Application for each type connection

(8) Cleaning procedure or reference to the cleaning procedure used

(9) Selection of the instrumentation for compatibility (materials, temperature, pressure, etc.) with the ambient environment and with the parameter being measured.

30.9 Drawing format.

a. Unless otherwise approved by the authorized government activity, a maximum of three sheets shall be allotted for single functioned system instrumentation and a maximum of ten sheets for instrumentation containing a multiple functioned system. A single functioned system is one that performs only one operation such as: expands/contracts a pressure elastic element, steps up/steps down the voltage, conditions one electrical signal, or winds/unwinds a bimetallic element. A multiple functioned system is one that contains two or more single functioned systems.

b. Each sheet shall be zoned.

c. Title block shall be included on each drawing sheet and shall include the following information:

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(1) Title, drawing number, and revision letter. Each sheet contain the same title, drawing number and revision letter.

(a) Title - The title shall consist of the name by which the instrumentation is known.

(b) Drawing number - The drawing number shall consist of alpha-numeric characters which may be separated by dashes or slashes. The total number of characters in the drawing number (including dashes and/or slashes) shall not exceed 15. Blank spaces are not permitted within the drawing number.

(c) Revision letter - The revision letter shall denote the latest approved version of the drawing. The revision letter of conformance verification drawing shall not be changed until all the changes under that revision have been accepted in writing by the authorized government activity (NAVSEA). No changes made to the conformance verification drawing shall be considered a revision until after the initial version of the conformance verification drawing has been accepted in writing by the authorized government activity. After initial submittal of the conformance verification drawing, no changes shall be made during the initial drawing review process unless the change is either requested by the authorized government activity or the change is documented in written correspondence by the contractor.

(2) Sheet _____ of _____

(3) Tolerance on dimensions for fractions, decimals, and angles. Units of the dimensions specified on the conformance verification drawing.

(4) Contractor acceptance block (appropriate signatures and dates)

(5) Federal supply code for manufacturer (FSCM)

(6) Scale

(7) Reference drawings

(8) Manufacturer's name and address

(9) Drawing size

d. Revision block. The revision block shall be included on each sheet of the conformance verification drawing and shall contain the following information in tabular form:

(1) Revision letter

(2) Description of revision

(3) Acceptance letter serial number and originator identification

(4) Acceptance date

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e. Form. Sheet size and format not specified herein shall be in accordance with ANSI Y14.1.

f. Classification designations. No government security classification designation such as confidential or secret shall appear on the conformance verification drawing unless a particular classification is specified by the government.

40. CONFORMANCE VERIFICATION DRAWING ACCEPTANCE

40.1 Acceptance. Acceptance shall be granted by the authorized government activity only after the conformance verification drawing is found to meet all the requirements specified in 20 through 40.1.

50.1 Acquisition document instructions. The acquisition document should contain provisions that address submission, review, extension, disapproval, default, acceptance, and waiver of conformance verification drawings in addition to the effects on the delivery schedule due to delays in conformance verification drawing acceptance. NAVSEA shall be designated as the activity that accepts or disapproves the conformance verification drawing.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-G-24659A(SH)	2. DOCUMENT DATE (YYMMDD) 27 SEPTEMBER 1993

3. DOCUMENT TITLE
GAUGE, DIFFERENTIAL, DIAL INDICATING

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME TECHNICAL POINT OF CONTACT (TPOC): D. STRAWSER, SEA 03Z44 NAVAL SEA SYSTEMS COMMAND	b. TELEPHONE (Include Area Code) (1) Commercial (TPOC) 703-602-6793	(2) AUTOVON 8-332-6793
c. ADDRESS (Include Zip Code) SEA 03Q42 2531 JEFFERSON DAVIS HWY. ARLINGTON, VA 22242-5160	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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